

How should I investigate data quality?

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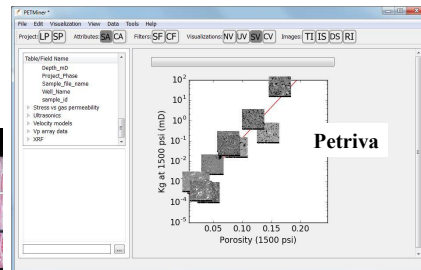
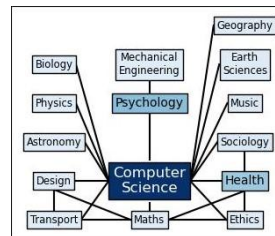
- Inter-disciplinary background
- Expert in data visualization and data quality
 - Hospital and GP data
 - Radio astronomy
 - Supermarkets & convenience stores
 - Adult social care
 - Petrophysics
 - Train doors!

Data quality

- Open source
- Method & PyPI package
- Training



Leeds Virtual Microscope

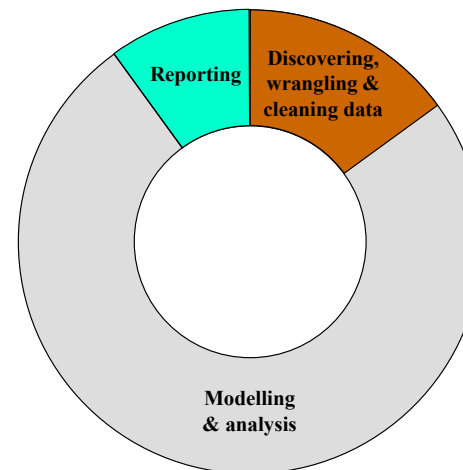


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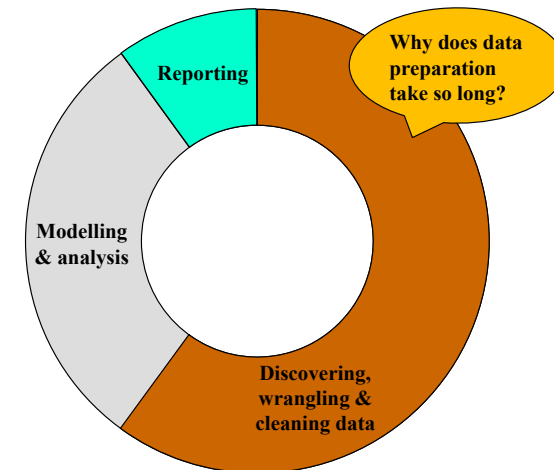
Overview

- Introduction
 - A data science workflow
 - Data quality taxonomy
 - What do data scientists want?
- Practitioner's guide
- Software (vizdataquality PyPI package)
- Benefits
 - Save time
 - Reduce re-work
 - Improve results quality

Workflow



Idealised data science project



Real project



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Lack of training / Ad hoc approaches

- **Typical data scientist**
 - Has little formal training in data quality
 - Cleans data in ad-hoc manner, following undocumented process
 - Only duplicate, missing value & missing record checks universally performed
 - Regrets not doing more
- **Wants**
 - To know what constitutes good practice
 - A rulebook (or formalised process) to follow

Practitioner's guide
<https://doi.org/10.5518/1481>



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What can go wrong?

- **Missing data**
 - Recording practices changed, risk of bias, etc.
 - Ethnicity data not recorded for two-thirds of perpetrators ... not robust enough to support national-level conclusions about offenders [Baroness Casey report]
- **Coverage** → discover some sensors are missing
- **Superfluous data** → cause noise in analysis
- **Value length change** → cannot track encrypted patient identifiers
- **Correct but misleading**
 - Spike in transactions before shop opened (picking to deliver online orders)

Everything that can be wrong will be wrong, on some occasion ... sooner than you think



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Solution

- **What?**
- **When?**
- **How?**
- **Why?**

Practitioner's guide
<https://doi.org/10.5518/1481>

What: A taxonomy of data quality

TAXONOMY OF DATA QUALITY TYPES AND CHARACTERISTICS (Level 1-3)			
Type	Level 1	Level 2	Level 3
Accuracy (a,b,c,d,g) Completeness (a,b,c,d,g) Consistency (a,b,c,d,g) Coverage (a,b,c,d,g) Precision (a,b,c,d,g) Reliability (a,b,c,d,g)	Integrity (a,b,c,d,g)	Missing record (a,b,c,d,g) Missing data (a,b,c,d,g) Missing value (a,b,c,d,g)	Dummy entry (a,b,c,d,g) Item non-response (a,b,c,d,g) Semi-empty tuple (a,b,c,d,g) Zero value (a,b,c,d,g)
	Duplicates (a,b,c,d,g)	Exact duplicates (a,b,c,d,g) Duplicate headers (a,b,c,d,g) Duplicate records (a,b,c,d,g)	Complexity missing variable* (a,b,c,d,g) Complexity missing header* (a,b,c,d,g) Missing column name* (a,b,c,d,g)
	Coverage (a,b,c,d,g)	Event coverage (a,b,c,d,g) Duplicate headers (a,b,c,d,g) Appropriate accuracy of data (a,b,c,d,g) Coverage representation (a,b,c,d,g) Relevance (a,b,c,d,g)	Rate of recording (a,b,c,d,g) Coding specificity (a,b,c,d,g) Relevant time interval (a,b,c,d,g)
	Accuracy (a,b,c,d,g) Completeness (a,b,c,d,g) Consistency (a,b,c,d,g) Coverage (a,b,c,d,g) Precision (a,b,c,d,g) Reliability (a,b,c,d,g)	Accuracy (a,b,c,d,g) Completeness (a,b,c,d,g) Consistency (a,b,c,d,g) Coverage (a,b,c,d,g) Precision (a,b,c,d,g) Reliability (a,b,c,d,g)	Accuracy (a,b,c,d,g) Completeness (a,b,c,d,g) Consistency (a,b,c,d,g) Coverage (a,b,c,d,g) Precision (a,b,c,d,g) Reliability (a,b,c,d,g)
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Dummy entry
 Item non-response
 Semi-empty tuple
 Zero value

Approximate duplicates

Ruddle. (2023). Proc. IVAPP.



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What: A practitioner's guide

<https://doi.org/10.5518/1481>

- 62 characterisation and data quality tasks

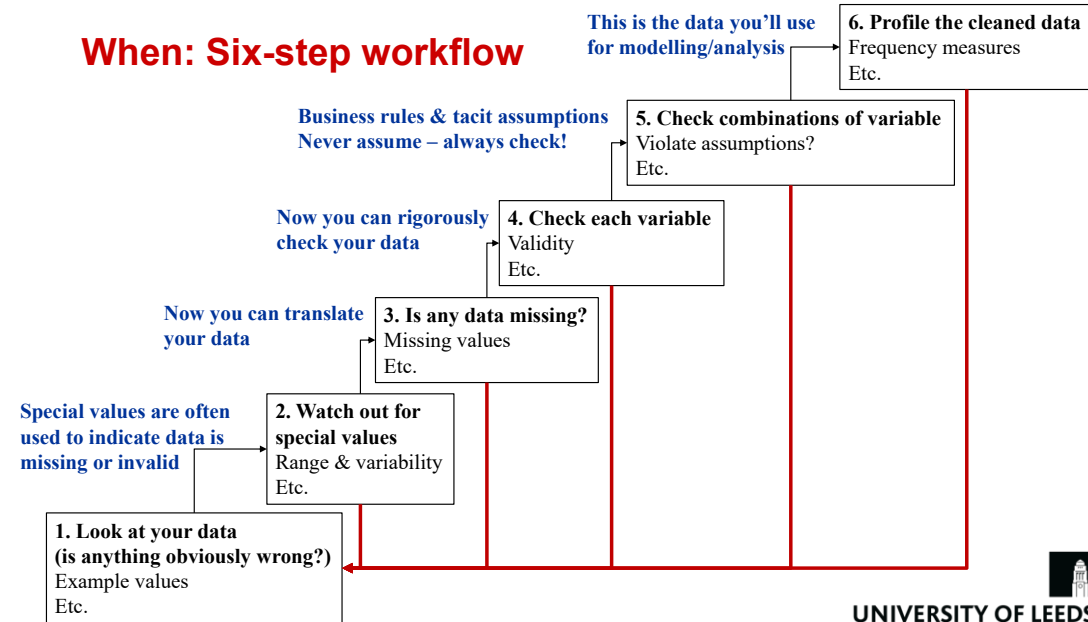
- Plain English
- Each task articulated as question(s) to answer about your data

Task	Question(s)
Example values	What does the data look like (print the first/last few rows)?
Range & variability	What is the minimum/maximum value? What are certain percentiles (e.g., 25th, median and 75th)?
Missing values	How many values are missing? Did you expect the variable to be complete, but some values are missing?
Validity	Do any values contain characters or symbols that should not be present?
Violate assumptions?	What assumptions have you made about variables (are they true)?
Frequency measures	What is the distribution of the values? How many times does each value occur?



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When: Six-step workflow



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How: vizdataquality

- Python package

- <https://pypi.org/project/vizdataquality/>
- 16 example Jupyter notebooks
(<https://github.com/royruddle/vizdataquality>)

- Data profiling calculations & visualizations

- Uses Pandas and Matplotlib
- If necessary, customise visualizations with **kwargs

- Use the six-step workflow to do your investigations

- Clean your data & create a report at the same time

- Output text, figures & tables as webpage, Latex or for Word



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Leeds parking fines

- Open data

- <https://datamillnorth.org/dataset/v8ggw/off-street-parking-fines>
- Quarterly datasets, Apr 2013 to Mar 2022

- Example dataset: Jan – Mar 2019

- See vizdataquality “Workflow (parking fines)” notebook



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Step 1: Look at your data (is anything obviously wrong?)

```
df = pd.read_csv('Quarter 4 201819.csv')
```

```
print(df.iloc[-1])
```

PCN	LS08541469	
ISSUED	31/03/2019	← What time?
LOCATION	WEST STREET CP - CITY	
CONTRAVENTION	83 WITHOUT DISPLAYING A VALID TICKET	
FINE		→ 50
Last Pay Date	31/03/2019	← Eh? → Illogical
Total Paid	0.0	
Balance	0.0	
Unnamed: 8	NaN	



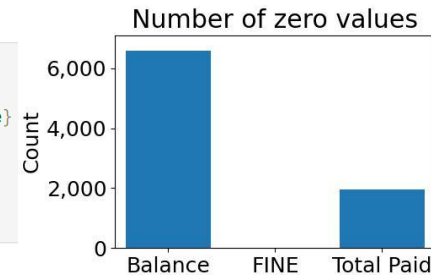
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Step 2: Watch out for special values

• Watch out!

- Certain values may have a special meaning (e.g., 999999)
- Some systems use '0' as a default value

```
stats = vdqc.calc(df)
cols = ['Balance', 'FINE', 'Total Paid']
fig_kw = {'size_inches': (5, 3.25), 'constrained_layout': True}
ax_kw = {'title': 'Number of zero values', 'xlabel': None,
        'ylim': (0, num_rows), 'ylabel': 'Count'}
vdqp.scalar_bar(stats.loc[cols]['Number of zero values'],
                fig_kw=fig_kw, ax_kw=ax_kw)
```



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Step 3: Is any data missing?

• Missing values

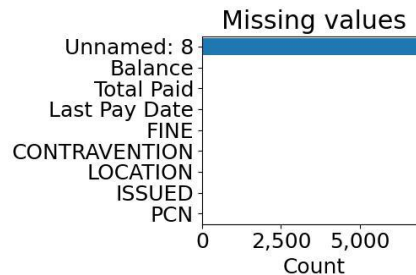
- Special values?
- Zeros?
- Invalid values?

• What is the extra “Unnamed: 8” variable?

```
fig_kw = {'size_inches': (5, 3.25), 'constrained_layout': True}
ax_kw = {'title': 'Missing values', 'ylabel': None, 'xlim': (0, num_rows),
        'xlabel': 'Count'}
vdqp.scalar_bar(stats['Number of missing values'], vert=False,
                fig_kw=fig_kw, ax_kw=ax_kw)
```

• Spatial coverage

• Temporal coverage



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Step 4: Check each variable

• Uniqueness

• Date & time distributions

• Value lengths

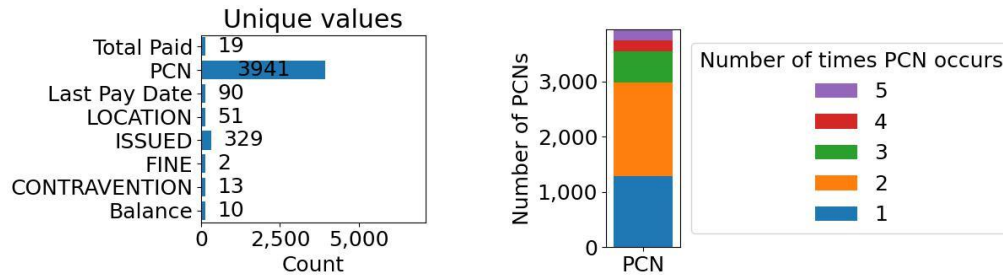
- NB: That's just some of the tasks (see Practitioner's guide <https://doi.org/10.5518/1481>)



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Step 4: Uniqueness

```
fig_kw = {'size_inches': (5, 3.25), 'constrained_layout': True}
ax_kw = {'title': 'Unique values', 'xlim': (0, num_rows), 'xlabel': 'Count',
        'ylabel': None}
vdqp.scalar_bar(stats['Number of unique values'], perceptual_threshold=0.02,
                vert=False, datalabels=True, fig_kw=fig_kw, ax_kw=ax_kw)
```



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Step 4: So how much was paid in total?

- Some PCNs are paid in parts, not in one go
 - So you will get the wrong answer if you just sum the Total Paid

```
# What is the sum of the Total Paid column?
print('Sum of the Total Paid column =', df['Total Paid'].sum())

# But what was the total amount paid?
print('Total actually paid =', df.groupby('PCN')['Total Paid'].max().sum())
```

Sum of the Total Paid column = 191106.02

Total actually paid = 106806.93



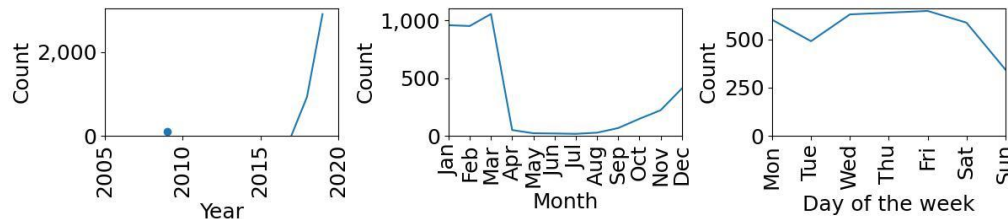
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Step 4: Dates

NB: Each PCN is only counted once!

- Check frequency distribution at multiple levels of detail

```
fig_kw = {'size_inches': (12, 3), 'constrained_layout': True}
ax_kw = {'title': 'Distribution of ISSUED values'}
components = ['year', 'month', 'dayofweek']
xrot = [90, 90, 90]
vdqp.plotgrid('datetime distribution', df_noduplicates['ISSUED'], num_rows=1,
              xlabel_rotate=xrot, fig_kw=fig_kw, ax_kw=ax_kw, components=components)
```

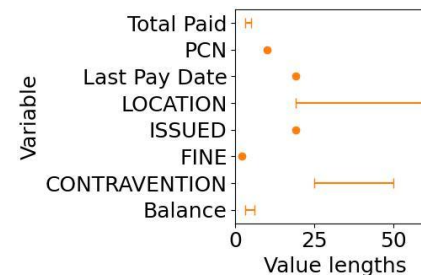


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Step 4: Value lengths

- How many characters are in the values of each variable?
 - Big ranges often indicate data quality issues

```
fig_kw = {'size_inches': (3, 2)}
ax_kw = {'xlim': 0}
vdqp.dot_whisker(stats['Value lengths'].sort_index(), vert=False,
                 fig_kw=fig_kw, ax_kw=ax_kw)
```



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Step 4: Big value length ranges

```
cols = ['LOCATION', 'CONTRAVENTION']
df_examples = vdqc.get_value_lengths_examples(df[cols])

for column in df_examples.columns[1:]:
    print(column)
    for row in df_examples[['Examples', column]].itertuples(index=False):
        print(row[0].rjust(14) + ': ', "" + row[1] + "")
```

LOCATION

Shortest value: 'CASTLE ST CP - CITY'

Median value: 'KIRKSTALL LEISURE CENTRE - CP'

Longest value: 'BARLEY HILL ROAD CP - GARFORTH'

CONTRAVENTION

Shortest value: '73 PARKED WITHOUT PAYMENT'

Median value: '84 ADDITIONAL PAYMENT MADE TO EXTEND TIME'

Longest value: '81 IN A RESTRICTED AREA IN A CAR PARK'



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Step 5: Check combinations of variables

- Variety of tasks (see Practitioner's guide <https://doi.org/10.5518/1481>)
- E.g.,
 - What assumptions have you made?
 - Do any sets of values violate rules?

Variables
PCN
ISSUED
LOCATION
CONTRAVENTION
FINE
Last Pay Date
Total Paid
Balance



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Step 5: Violate assumptions

- Why?

Fine – Total Paid – Balance (£)	Count
-116	3
-100	5
-56	4
-50	24
-43	18
-35	28
-33	729
-25	332
-10	1
0	785
25	3352
35	215
49.06	1
49.1	1
50	1559
70	51



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Step 6: Profile the cleaned data



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Workflow: Report

- Three formats
 - Text & images (manually import into Word)
 - Latex
 - HTML

Why?

- Do more in less time (know what to do; the tasks & questions)
 - Reduce time & cost (by avoiding re-work)
 - Understand any limitations of the data
 - Correct your assumptions
 - Improve your data
- } Improve results



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Summary

- Publicly available resources to help you rigorously investigate data quality and profile your data
 - A Practical Guide to Characterising Data & Investigating Data Quality (<https://doi.org/10.5518/1481>)
 - Python package: vizdataquality (<https://pypi.org/project/vizdataquality/>)
- One day CPD course
- YouTube “how to” videos (coming soon)
- Open data repository roll-out

Questions?

- Practitioner’s guide
<https://doi.org/10.5518/1481>
- vizdataquality
<https://pypi.org/project/vizdataquality/>
- For these slides, see
<https://github.com/rovruddle/tutorials-and-talks/blob/main/README.md#talks-about-data-quality>



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